Simulation-Based Wrist Exoskeleton Assistance after Stroke: Optimal Control vs. Reinforcement Learning

P2: Literature Map, Method Outline and Baseline Simulation

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P2 at a Glance:

1 **Literature map** — Done

19 sources in 4 clusters: clinical problem, wrist devices, musculoskeletal optimal control, learning-based control

2 **Method outline** — Done

Two-route pipeline (Moco plan and MyoSuite RL), KPIs K1–K4 made measurable

3 **Baseline simulation** — Fallback model

Moco cup lift-and-hold converges; muscles do all the work

4 **Bibliography** — Done

All references checked online; list on the last slides

0.02 %

peak reserve torque vs. gravity torque in the baseline (limit 10 %)

3.4 s

refined Moco solve (60 mesh intervals, 24 iterations)

≈ 10.7 %

extensor strength below which the task fails (K1 preview)

Numbers from the planar fallback model; MoBL-ARMS pending.

Literature Map:

Clinical problem (why)

- Nijland 2010: early finger extension predicts arm recovery after stroke
- Wrist drop: weak extensors, hand falls under gravity

Wrist assist devices (what exists)

- Hussain 2020: review of wrist rehabilitation robots
- Lambelet 2017, 2020: eWrist, 1-DoF wrist exo, < 500 g
- KhalilianMotamed Bonab 2024: exosuit designed in OpenSim

Our study

Size wrist assistance for a weakened wrist (spring or motor?) and compare a Moco plan with an RL policy

Optimal control (Route A)

- Delp 2007, Seth 2018: OpenSim
- Dembia 2020: OpenSim Moco
- Holzbaur 2005, Saul 2015: MoBL-ARMS
- De Groote 2016: muscle model

Learning-based control (Route B)

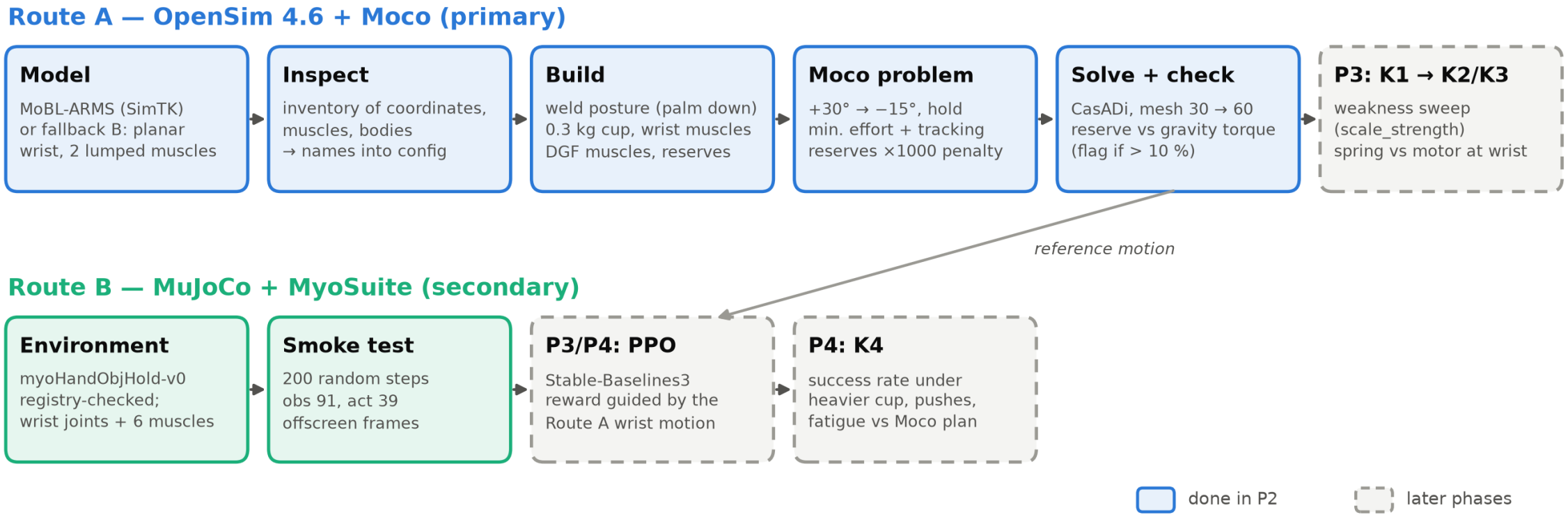
- Todorov 2012: MuJoCo
- Caggiano 2022, Wang 2022: MyoSuite, MyoSim
- Schulman 2017: PPO
- Luo 2024, Chavarrías 2025: RL exo control

Gap: no study we found sizes wrist assistance for a weakened post-stroke wrist, or compares an optimal plan with a learned controller.

From Literature to Our Pipeline:

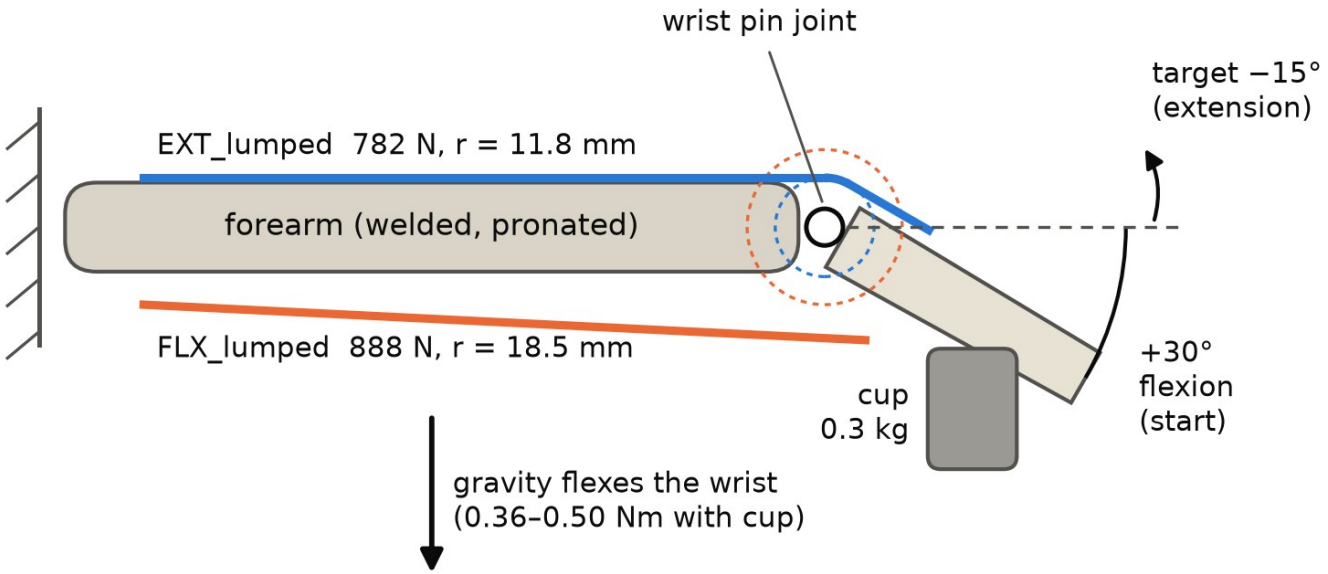
Source	What it gives us	Where we use it
Holzbaur 2005; Saul 2015	MoBL-ARMS upper-limb model and muscle parameters	Route A model (pending SimTK access)
Dembia 2020 (Moco)	Direct-collocation optimal control	Baseline solve, K1–K3 studies
De Groote 2016	Fast, robust muscle formulation	DeGrooteFregly muscles in every solve
Wang 2022; Caggiano 2022	MyoHand model (from MoBL-ARMS) and MyoSuite tasks	Fallback muscle forces / moment arms; Route B environment
Lambelet 2017, 2020; Hussain 2020	A 1-DoF wrist flexion/extension exo is wearable and realistic	Assist modelled as torque at the wrist axis
Schulman 2017; Raffin 2021	PPO and Stable-Baselines3	Route B policy training (P3/P4)

Method Outline:



Solid boxes: built and tested in P2. Dashed boxes: P3/P4. The Moco wrist motion becomes the reference for the RL reward.

Simulation Model and Task:



- Forearm fixed, horizontal, palm down: gravity flexes the wrist (wrist drop)
- Hand 0.45 kg (Winter 2009) plus a 0.3 kg cup
- One lumped extensor (ECRL + ECRB + ECU) and one flexor (FCR + FCU)
- Forces and moment arms from MyoHand (derived from MoBL-ARMS)
- DeGroote–Fregly muscles, rigid tendons

Fallback B. MoBL-ARMS download was pending SimTK account approval. Same pipeline; swapping it in is a config change.

Moco Problem Formulation:

1 Task

- Wrist from +30° (dropped) to −15° (extended)
- 1.0 s; last 30 % is a hold
- Zero velocity at start and end

2 Objective

- Minimise muscle effort (Σ excitation²)
- Reserve torque penalised ×1000
- Soft tracking of a smooth move-then-hold path

3 Pass / fail checks

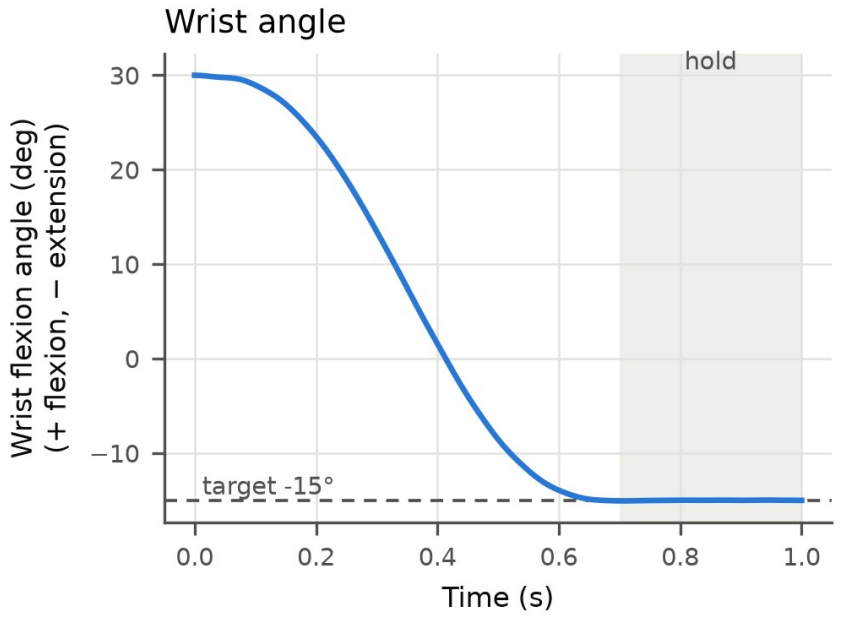
- Reserve ≤ 10 % of gravity torque
- Hold drift ≤ 2°
- Solver must converge

4 Solver

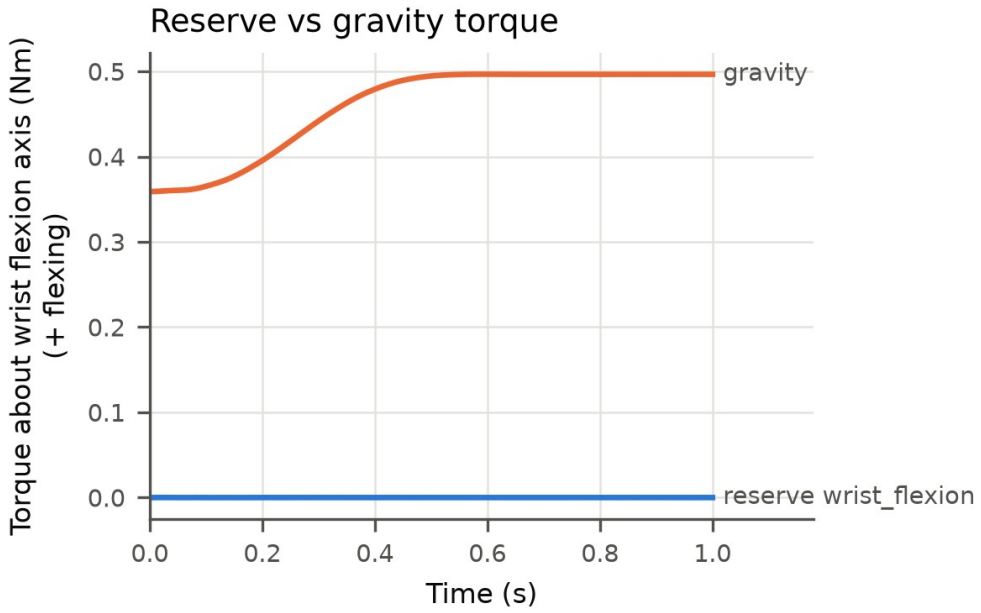
- Moco CasADi + IPOPT, Hermite–Simpson
- Coarse 30 → refined 60 mesh intervals
- Tolerances 1e-3

Baseline Result (100 % Strength):

Fallback B: planar 2-muscle wrist model



Fallback B: planar 2-muscle wrist model



Converged

3.4 s, 24 iterations on the 60-interval mesh

−15.00°

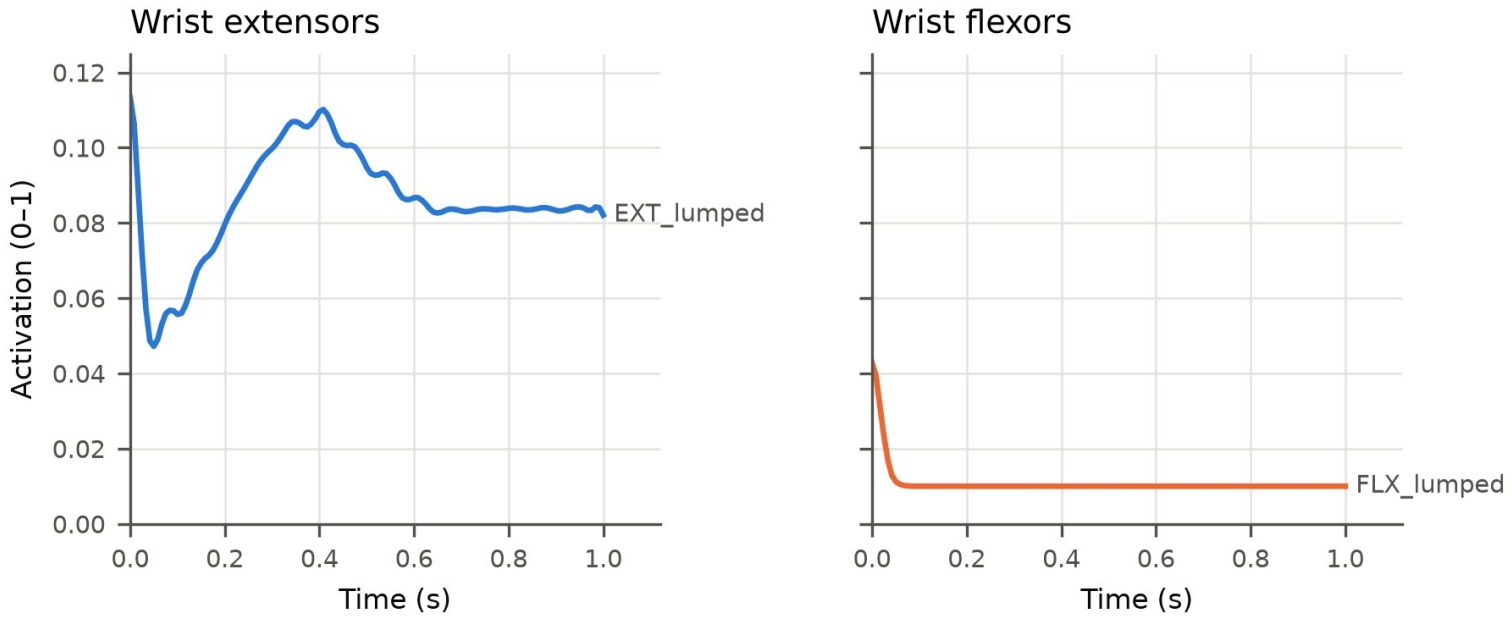
target reached; held within 0.06°

0.02 %

peak reserve / gravity torque: muscles do the work

Muscle Activity:

Fallback B: planar 2-muscle wrist model



Extensor works to lift and hold; flexor stays at the model's minimum activation (no co-contraction needed).

≈ 8 %

extensor activation holds the 0.3 kg cup level

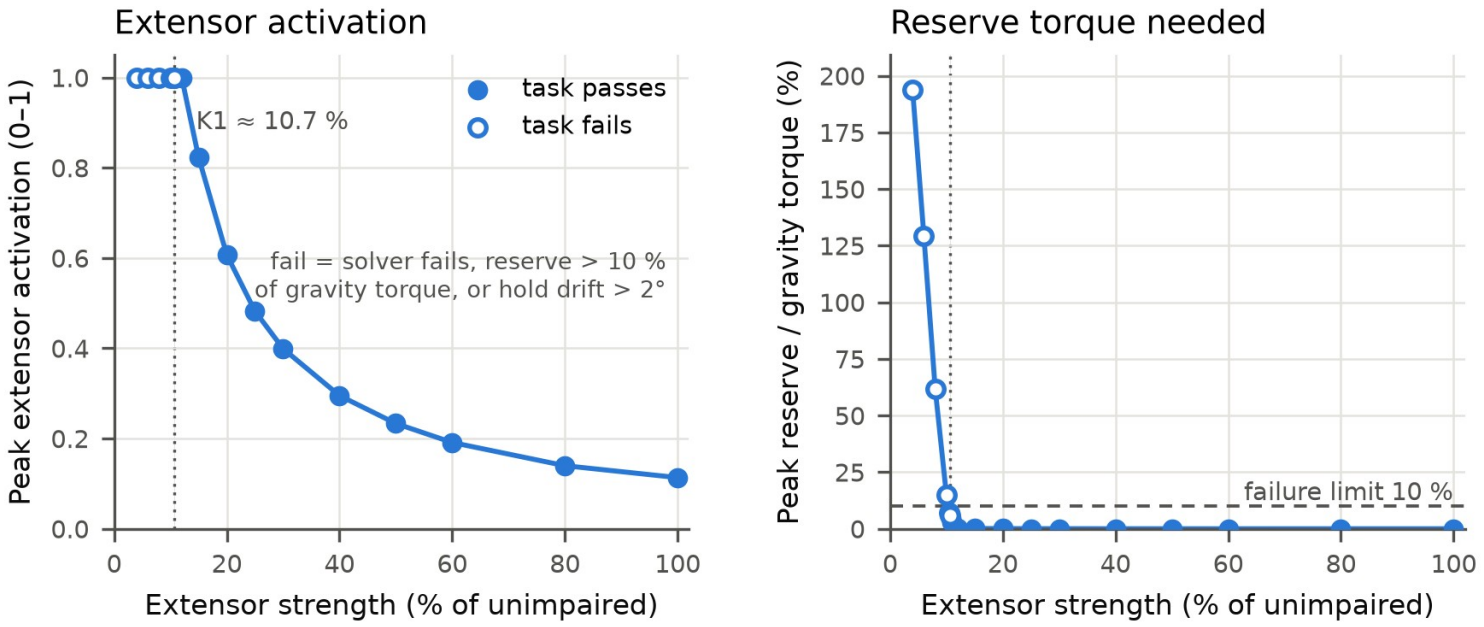
0.01

flexor activation (model minimum)

Starting point for K1: how much weaker can the extensor get?

K1 Preview: Weakness Sweep:

Fallback B: planar 2-muscle wrist model | preliminary P3 result



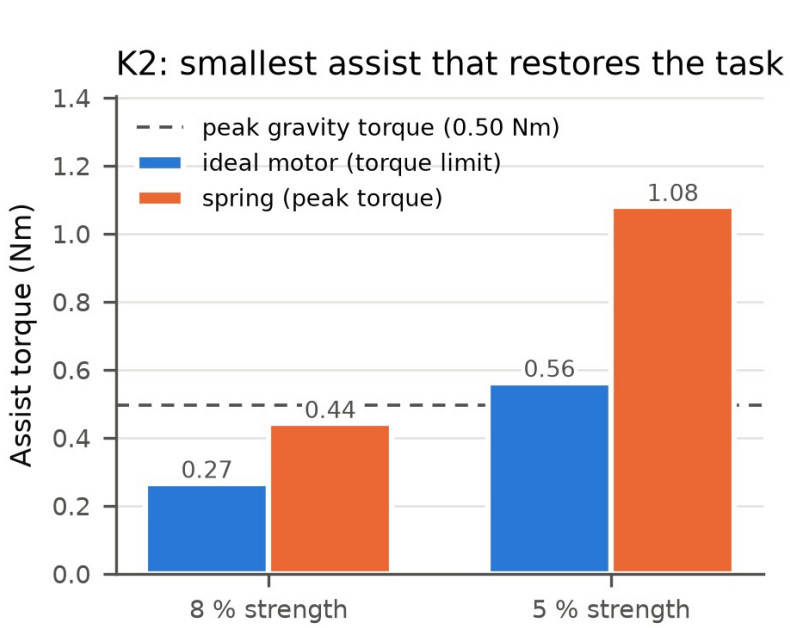
≈ 10.7 %

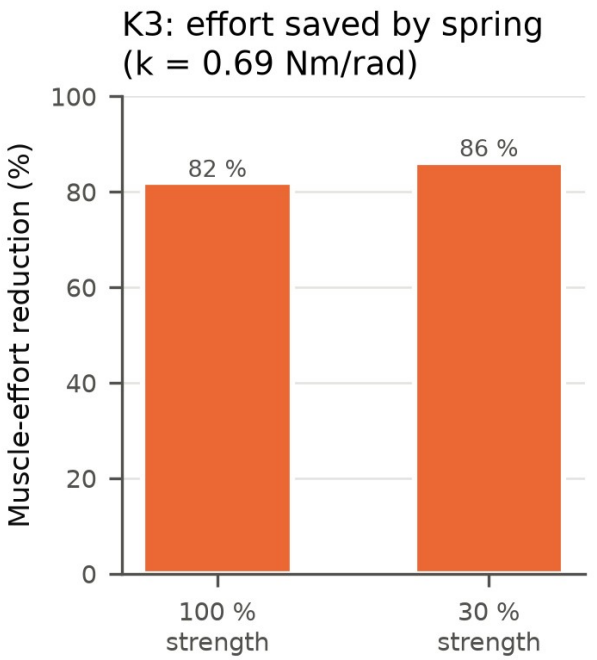
of unimpaired extensor strength: below this the cup task fails

- Activation saturates at 1.0 below ~12 %
- Threshold found by bisection: 10.62–10.75 %
- Preview on the fallback model; final K1 on MoBL-ARMS in P3

K2/K3 Preview: Motor vs. Spring:

Fallback B: planar 2-muscle wrist model | preliminary P3 result





0.27 Nm

motor that restores the task at 8 % strength (0.56 Nm at 5 %)

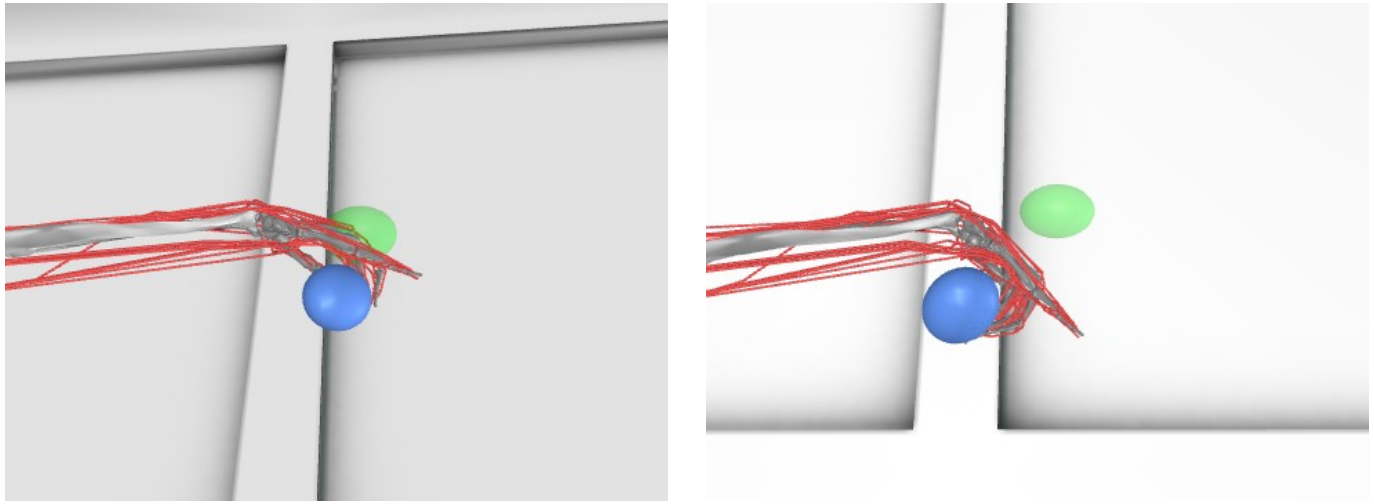
1.6–1.9×

spring peak torque vs. motor: a spring is enough, but must be stronger

82–86 %

less muscle effort with the spring when the task still works

Route B: MyoSuite Environment:



myoHandObjHoldFixed-v0 under random actions (offscreen render).

Check	Result
Registry	226 myo environments, 24 myoHand
Obs. / action	91 / 39 (muscles)
Wrist joints	flexion, deviation, pro_sup
Wrist muscles	ECRL, ECRB, ECU, FCR, FCU, PL
Smoke test	200 random steps, no errors

Route B plan (P3/P4)

- PPO (Stable-Baselines3); reward tracks the Moco wrist motion, keeps the cup level and penalises effort
- Fingers held in a fixed grasp first; policy outputs muscle (and device) commands
- K4: compare the RL policy with the Moco plan under a heavier cup, pushes and fatigue

Limitations and Validity:

Limitation	Effect	Plan
Planar fallback model (MoBL-ARMS pending)	Absolute numbers are model-specific	Re-run everything on MoBL-ARMS
Assumed fibre and tendon lengths	K1 value may shift	Sensitivity analysis in P3
Hold enforced by a soft tracking goal	Kinematics partly prescribed	Compare with a purely predictive variant
Spring rest angle and motor weight fixed	K2/K3 depend on these choices	Optimise them as Moco parameters
Generic adult, one 1 s task	Not a clinical prediction	State as relative findings (P1 ethics)

Next Steps (P3):

1

MoBL-ARMS swap

inventory, posture, baseline re-run

on SimTK approval

2

Final K1 + sensitivity

cup mass, speed, muscle parameters

P3 · 18 Nov

3

Spring vs. motor

stiffness and rest angle as Moco parameters

P3 · 2 Dec

4

Route B training

PPO in MyoSuite, Moco-guided reward

Weeks 48–50

5

Robustness (K4)

heavier cup, pushes, fatigue

P4 · 16 Dec

Already in place: sweep and assist scripts, tests and figures run unchanged on MoBL-ARMS (config switch).

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Thank You

Questions?

Repository:
github.com/Rationaliser/wrist-exo-assist-sim